

Healthcare Analysis Data report

Executive Summary:

This analysis highlights key financial and service-related insights across the healthcare system, focusing on patient costs, departmental trends, service types, satisfaction levels, and regional disparities .

Scope

1-Total Out Of Pocket.

2-Total Billing Amount.

3-Total Insurance Coverage.

4-Total Medication cost.

5-Total Treatment Cost.

6- Count Of Patient By Diagnosis ,Procedure, Service Type, Department.

7-Out Of Pocket By Diagnosis ,Procedure, Service Type, Department.

8- Out Of Pocket By Month.

9-Total Patient .

9- Patient Satisfaction Score by Number Of Patient.

10- Out Of Pocket By City & States Out Of Pocket By City & States.

Data Collection & Methodology:

- Data source: Healthcare Dataset (8 tables)(csv file).

- Tools Used : Power BI(power query , dax).

- Pre processing steps :add new measures by dax, replace values , change data type and create custom columns ,by power query.

- Data Model(table relations) : Visits (Fact table), departments - insurance - procedures - providers - cities - patients (Dim tables) - all relation 1---->M

Data Analysis:

» Total Out Of Pocket ,Total Billing Amount ,Total Insurance Coverage

,Total Medication cost ,Total Treatment Cost.

Insight:

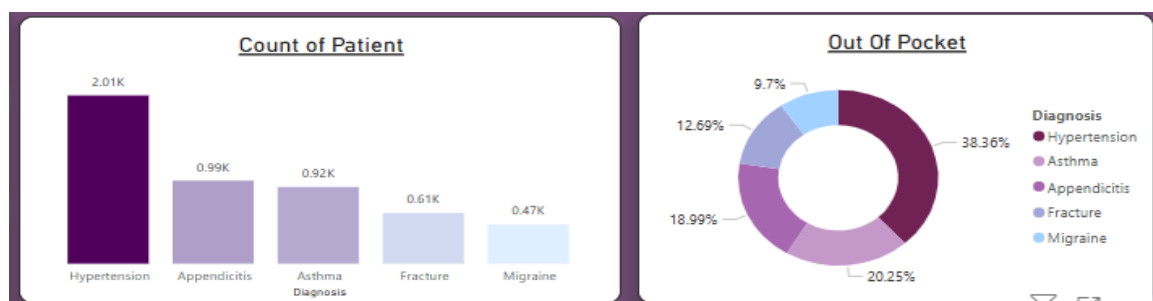
- **Total Out-of-Pocket: \$1.13M**, which is **~34%** of the **Total Billing Amount (\$3.36M)** ---> that means **1 out of every 3 dollars** billed was paid directly by patients.
- **Total Insurance Coverage: \$2.23M**, covering **~66%** of billing ---> This coverage level is moderate but not optimal
- **Medication Cost: \$546.5K** , which is **~16%** of total billing), **Treatment Cost: \$2.63M** , which is **~78%** of total billing) ---> Treatment dominates spending.
- **Medication + Treatment : \$3.18M** , **Out-of-Pocket = \$1.13M** ----> that mean Patients are covering **~35.5%** of core care (meds + treatments) themselves.(Insurance Coverage cover only **~70%** of the total care costs (meds + treatment)).

Out Of Pocket
1.13M
Billing Amount
3.36M
Insurance Coverage
2.23M
Medication Cost
546.04K
Treatment Cost
2.63M

Recommendations:

- 1- Negotiate Better Insurance Terms for High-Cost Treatments .

» Count Of Patient & Out Of Pocket By Diagnosis ?



insight:

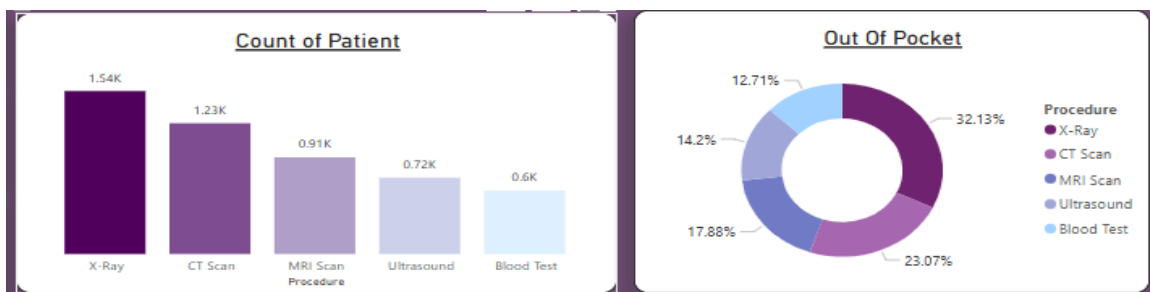
- **Hypertension Is a High-Burden Diagnosis**
 - With 2.01k patients, it's the most common diagnosis.
 - It also has the highest out-of-pocket share at 38.36%.
- **asthma (20.25%) and Appendicitis (18.99%)** have moderate volumes and non-trivial patient costs, despite being common, manageable conditions.

- Fracture and Migraine represent lower patient volumes and relatively low out-of-pocket percentages.

Recommendations:

- 1- to improve healthcare affordability, cost-reduction efforts should prioritize high-volume, high-burden chronic conditions like Hypertension and Asthma. These have the biggest impact on overall patient financial well-being and satisfaction.

» Count Of Patient & Out Of Pocket By Procedure .



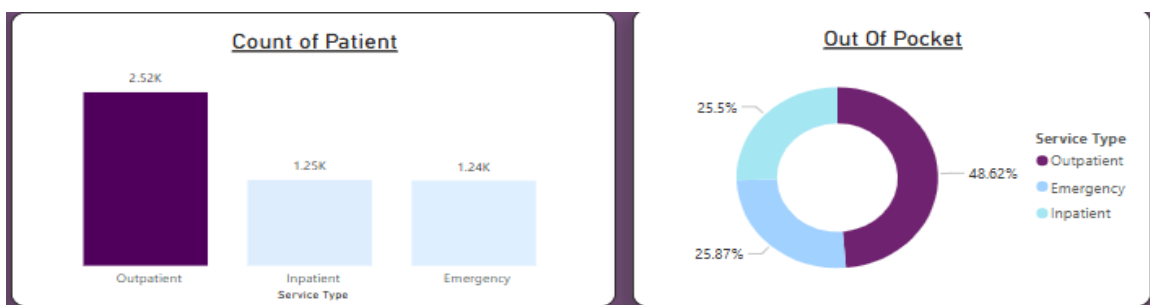
insight:

- X-Ray Has the Highest Volume and Cost Burden .
- CT Scans and MRI Scans Are Costlier but Better Covered.
- Ultrasounds and Blood Tests Have the Lowest Burdens.

Recommendations:

- improved insurance coverage for high-volume, high-cost diagnostics (X-rays).

» Count Of Patient & Out Of Pocket By Service Type .



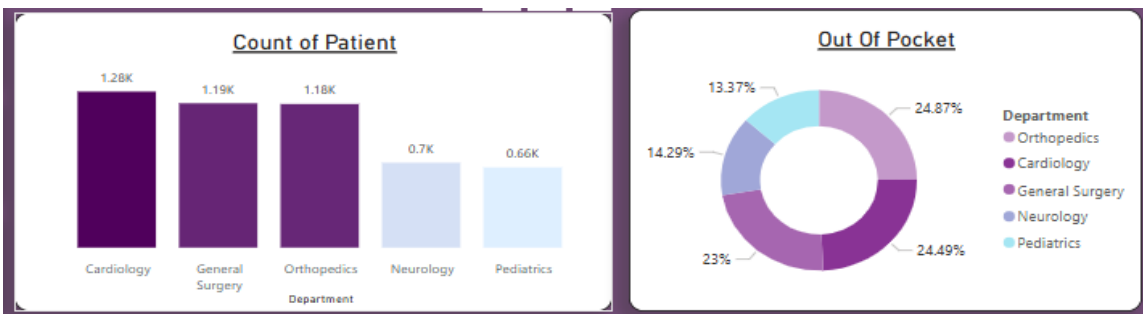
insight:

- Outpatient care—usually considered more accessible and lower cost—the most financially burdensome for patients. This might include regular visits, diagnostics, minor treatments, or therapy, which often fall outside major insurance coverages or under high co-pays.
- Despite being higher cost and more acute, emergency (25.87%) and inpatient care (25.5%) are better covered by insurance.

Recommendations:

- Review Insurance Plans for Outpatient Coverage
- Work with insurers to reduce co-pays and improve coverage for routine visits, minor procedures, and follow-ups.

»Count Of Patient & Out Of Pocket By Department.



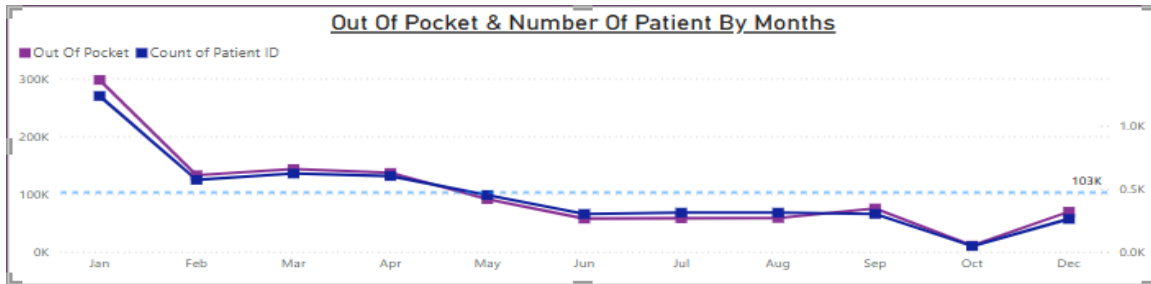
insight:

- Orthopedics, Cardiology & General Surgery High Cost and High Volume.
- Neurology and Pediatrics Have Lower Burdens.

Recommendations:

- 1- Offer pre-surgery financial counseling and payment plans in Orthopedics and General Surgery to manage large bills.
- 2- Explore preventive care programs in Cardiology to reduce downstream costs and patient burden.
- 3- Continue supporting low out-of-pocket strategies in Pediatrics and Neurology, especially if tied to government or public health programs.

» Out Of Pocket By Month?



insight:

- January: Highest out-of-pocket AND highest patient count — **strong start-of-year** surge(The steep spike in January out-of-pocket costs is disproportionately high, even accounting for patient volume).
- Feb–May: **Stable** patient count and costs.
- Jun–Sep: **Decline** in both patients and out-of-pocket.
- October: **Lowest** patient count and out-of-pocket — **significant drop**.
- December: **Mild rebound** in both.

Recommendations:

Run discounted checkup or screening drives to increase engagement when clinics are underused

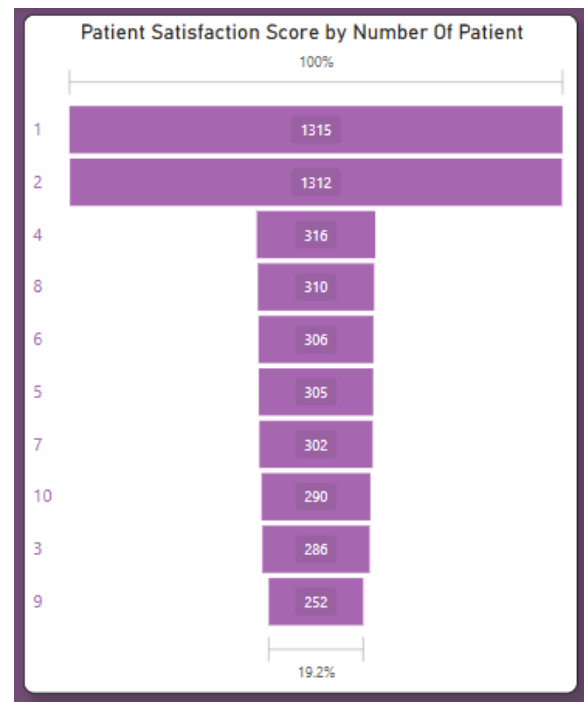
» Patient Satisfaction Score by Number Of Patient?

insight:

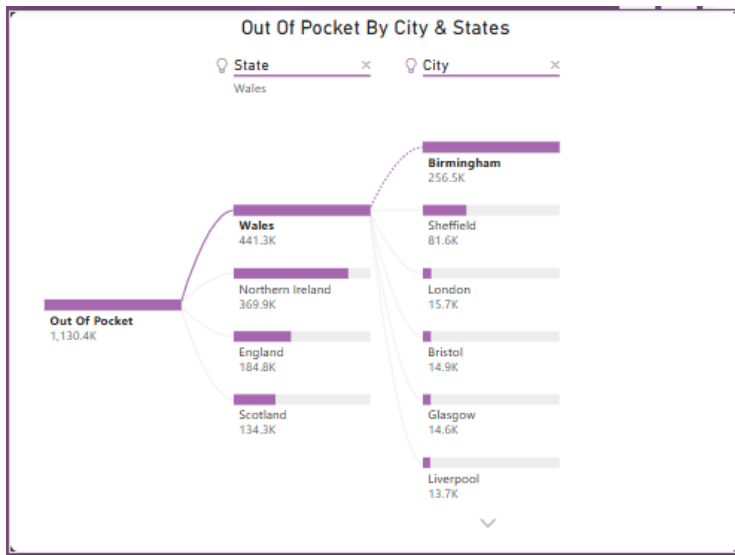
- Nearly 52% of patients gave the lowest satisfaction ratings.
- Scores 3 to 10 drop off dramatically — with score 3 at only 286 patients, and score 10 (the highest) at 290.

Recommendations:

- 1- Use surveys or calls to gather feedback and improve operations.
- 2- Promote Feedback from Satisfied Patients (Scores 8–10).



Out Of Pocket By City & States



insight:

- Patients in Wales experience the highest financial burden compared to other regions.
- Scotland Has the Lowest Out-of-Pocket Costs.

Recommendations:

- 1- Explore opportunities to replicate Scotland's cost-management strategies in higher-cost regions like Wales.

“By implementing these data-driven insights, we aim to enhance healthcare accessibility, reduce patient financial burden, and improve overall service satisfaction. We look forward to further discussions on refining these strategies and executing the next phase of care optimization and system-wide improvement.”

Prepared by:

- Data Analyst : Doha Masoud.
- Email: dohamasoud25@gmail.com